ABDULAHI OLAJIDE AFOLABI

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EDUCATION

B.Eng. (Hons) in Electrical and Electronics Engineering **Expected Jul 2026**
Federal University Oye-Ekiti, Nigeria

- **CGPA:** 4.43 / 5.00 (Second Class Upper Division)
- **Relevant Coursework:** Control Systems, Power Electronics, Power System Principles, Industrial Electronics, Microprocessors Hardware and Software Techniques
- **Capstone Project:** Development Of A Voice-Controlled Wheelchair To Improve Mobility, Independence, And Accessibility For Persons With Mobility Disabilities

National Diploma in Electrical and Electronic Engineering **Dec 2020**
Federal Polytechnic Offa, Nigeria

- **CGPA:** 3.54 / 4.00 **(Distinction) | Award:** Rector's Honours List.

HONOURS & AWARDS

- **Scholarship Recipient,** Sterling Bank Beyond Education Initiative **Sep 2025**
- **Scholarship Recipient,** Power Learn Project (Software Development) **Jul 2025**
- **Best Direct Entry Student,** Natl. Assoc. of Elec. & Electron. Eng. Students **Jul 2025**
- **Legislator of the Year,** FUOYE Student Union Senate Council **Jun 2024**
- **Scholarship Recipient,** NANS x PressPayNG (National Merit) **Dec 2023**
- **Second Best Graduating Student,** Elec. & Electron. Dept., Fed. Poly Offa **Dec 2020**

RESEARCH INTERESTS

- **Power System Monitoring & Analytics:** SCADA-based monitoring, data-driven optimisation, and fault diagnostics
- **Intelligent Control & Optimisation:** Fuzzy logic, adaptive controllers, and predictive control strategies for energy systems
- **Smart Grid & Renewable Energy Systems:** Microgrid modelling, stability analysis, and renewable integration

RESEARCH EXPERIENCE

Capstone Researcher | Voice-Controlled Smart Wheelchair **Sep 2025 – Present**
Federal University Oye-Ekiti | Department of Electrical and Electronics Engineering

- Designed hybrid distributed control system, demonstrating scalable real-time control applicable to power system and microgrid management
- Developed triple-redundancy safety algorithms (ultrasonic obstacle avoidance + manual override), reflecting fail-safe system design principles
- Applied signal processing (acoustic echo cancellation) achieving 95% recognition accuracy under high-noise conditions, transferable to predictive energy monitoring
- Validated actuator topology for thermal and load stability under dynamic conditions, demonstrating adaptive control relevant to distributed energy systems

Research Assistant | Hybrid Solar Energy Systems **Jul 2020 – Dec 2020**
Federal Polytechnic Offa | Department of Electrical and Electronics Engineering

- Designed a 2 kVA hybrid solar energy system using load profile analysis to optimize PV array sizing and battery bank capacity under variable demand conditions
- Engineered an Automatic Transfer Switch (ATS) based on relay logic, achieving sub-second (<1 s) transition latency between grid and inverter supply
- Implemented protection mechanisms including under-voltage lockout and overload protection to enhance battery lifecycle and system reliability
- Evaluated system performance under simulated outage scenarios, validating supply continuity and transient response stability

CERTIFICATIONS

- **IoT and Cloud Computing,** Yonsei University **May 2025**
- **MATLAB,** NIPES **May 2025**
- **AI for Software Engineering,** Power Learn Project **Mar 2026**

WORKSHOPS & CONFERENCES

- **IEEE HOT Interconnects Symposium** | Virtual **Aug 2025**
- **FUOYE 3rd International Engineering Conference** | Oye-Ekiti, Nigeria **May 2025**

PROFESSIONAL ENGINEERING EXPERIENCE

Engineering Trainee | Transmission Maintenance & System Monitoring **Mar 2025 – Sep 2025**

Transmission Company of Nigeria

- Monitored real-time grid variables via SCADA telemetry, analysing disturbance patterns and transmission line fault events to support network stability
- Conducted diagnostic and preventive maintenance on high-voltage transformers (330 KV / 132 KV), focusing on insulation integrity and operational safety
- Participated in fault response and system restoration procedures, reinforcing understanding of protection coordination and grid resilience
- Co-developed a prototype Transformer Ratio Test Kit, reducing manual testing time by 30% while improving turn-ratio verification accuracy

Engineering Intern **Jul 2019 – Jan 2020**

Ogo - Oluwa Engineering Workshop

- Designed and assembled an Arduino-based automatic 12 V battery charger using a voltage comparator circuit and relay switching logic, halting charging at full-charge threshold and resuming at a low-voltage setpoint to prevent overcharge and deep discharge.
- Installed and commissioned off-grid solar power systems, including PV panel mounting, charge controller wiring, battery bank connection, and inverter output configuration for residential loads.
- Wired and installed automatic changeover switches (ACS) for seamless transition between mains supply and inverter output, including contactor selection, neutral switching, and load-side protection.
- Diagnosed and repaired faults in solar charge controllers and inverter units, identifying failed MOSFETs, burnt shunt resistors, and degraded battery cells through component-level testing.

LEADERSHIP & COMMUNITY IMPACT

Sustainability & Digital Literacy Mobiliser **May 2025 – Nov 2025**

UNICEF x YOMA Skill-Up Programme

- Executed community-level sustainability initiatives including coordinated tree-planting and recycling campaigns, promoting environmental stewardship
- Facilitated onboarding of 150 students into digital learning platforms, improving access to professional development resources and digital inclusion

Senator & Policy Implementation Chairman **Jun 2023 – Present**

Students' Union Government (SUG), Federal University Oye-Ekiti Ekiti

- Authored campus safety policies and oversaw their implementation, ensuring student motions and policies were effectively executed
- Led oversight of policy implementation workflows, coordinating between student bodies and university administration

Academic Mentor & Peer Lead **Jan 2019 – Present**
Federal University Oye-Ekiti & Federal Polytechnic Offa

- Delivered structured free peer-learning sessions and tutorials for 200 undergraduate engineering students, simplifying complex engineering concepts
- Mentored 50 National Diploma students, contributing to measurable academic outcomes (5 Distinctions, 20 Upper Credits)

PROFESSIONAL MEMBERSHIPS

- Member, International Association of Engineers (IAENG) and its Societies
- Student Member, National Institute of Professional Engineers and Scientists (NIPES)
- Member, National Association of Electrical and Electronics Engineering Students

TECHNICAL SKILLS

- **Embedded Systems:** Arduino (AVR), Raspberry Pi (ARM Cortex), Sensor Fusion (Ultrasonic, Microphone Arrays), Motor Control (H-Bridge/BTS7960, PWM), ADC/DAC, Interrupt Handling
- **Circuit & PCB:** Proteus Design Suite, Transformer Vector Group Testing, PCB Layout & Fabrication
- **Programming:** Python (NumPy, Pandas), MATLAB/Simulink, C/C++, Octave
- **Industrial Automation:** Relay Logic, SCADA Monitoring, Automatic Transfer Switches

LANGUAGES

- **English:** Full Professional Proficiency (Medium of Instruction for all degrees)
- **Yoruba:** Native

REFEREES

1. Prof. Olaitan Akinsanmi
Professor, Department of Electrical & Electronics Engineering
Federal University Oye-Ekiti (FUOYE)
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